



VIDYARTHI VIGYAN MANTHAN 2026-27

SYLLABUS (CLASS - IX)

CLASS IX

PHYSICS

- 1. Motion:** Distance and displacement, velocity; uniform and non-uniform motion along a straight line; acceleration, distance-time and velocity-time graphs for uniform motion and uniformly accelerated motion, elementary idea of uniform circular motion.
- 2. Force and Laws of Motion:** Force and Motion, Newton's Laws of Motion, Action and Reaction forces, Inertia of a body, Inertia and mass, Momentum, Force and Acceleration. An elementary idea of conservation of Momentum.
- 3. Gravitation:** Gravitation; Universal Law of Gravitation, Force of Gravitation of the Earth (gravity), Acceleration due to Gravity; Mass and Weight; Free fall.
- 4. Floatation:** Thrust and Pressure. Archimedes' Principle; Buoyancy.
- 5. Work and Energy:** Work done by a Force, Energy, power; Kinetic and Potential energy; Law of conservation of energy (excluding commercial unit of Energy).
- 6. Sound:** Nature of sound and its propagation in various media, speed of sound, range of hearing in humans; ultrasound; reflection of sound; echo.

CHEMISTRY

- Exploring Mixtures and their Separation.
- Journey Inside the Atom.
- Atomic Foundations of matter.

BIOLOGY

- Cell structure and function, Cell division
- Animal tissues
- Plant tissues
- Reproduction: How Life Continues
- Patterns in Life: Diversity and Classification
- Biogeochemical cycle and human intervention

MATHEMATICS

- 1. Pure Arithmetic:** Rational and Irrational Numbers, Surds, Nested Radical.



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- 2. Algebra:** Algebraic Expansions, Factorization, Simultaneous Linear Equations and Inequalities, Factor Theorem and Remainder Theorem, logarithms, Set Theory, Permutation and Combination, AP and GP
- 3. Geometry:** Euclid Geometry, lines and angles, Triangles and its Properties, Quadrilaterals, Circles – incircle, circumcircle and excircle, centroid theorem, area of Pentagon and Hexagon, Midpoint theorem
- 4. Mensuration:** Area and perimeter of various shapes, like 2-D shapes, such as a triangle, a quadrilateral, a pentagon, and a hexagon, Heron Formula and Brahmagupta's rule of cyclic quadrilateral.
- 5. Statistics:** Introduction, collection, and presentation of data, Graphical representation, Mean and median of ungrouped data, Basic Probability
- 6. Coordinate Geometry:** Distance formula, Section formula.

LOGIC AND REASONING

Pattern Recognition: Number Series, Alphabet Series, Image Series, Odd One Out, Analogy, Figure Matrix & Matrix Reasoning, Pattern-based Puzzles, Coding-Decoding, Logical Sequence of Words, meaningful word formation.

Analytical Reasoning: Blood Relations, Direction Sense, Order & Ranking, Seating Arrangement, Grid Puzzles, Games & Tournaments, Scheduling Problems, Clocks & Calendar, Sequencing & Ordering Problems, Sudoku.

Visual Reasoning: Mirror Images, Water Images, Rotation, Paper Folding, Paper Cutting, Figure Counting, Embedded Figures, Shape Construction, Spatial Visualization, 3D visualization, Cubes, Dice & Cube Nets, Best Fit, Symmetry.

Mathematical Reasoning: Mathematical Operations, Number Puzzles, Cryptarithmic (Alphametics) puzzles, Binary Logic, Logical Inequalities, Logical Venn Diagrams, Word Problems.

Critical Thinking: Data Sufficiency, Decision Making, Logical Connectives, Syllogism, Input Output series, Critical Path Analysis, Scheduling Problems.

Emerging Reasoning Skills: Constraint Satisfaction Problems, Follow the Algorithm, Algorithm Output Prediction, Scientific Reasoning, Logical Puzzle Sets, Optimization, Geometrical reasoning.

LIFE STORY OF INDIAN SCIENTIST(S)

Asima Chatterjee : Unlocking Nature's Chemistry (All Chapters)



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INDIAN CONTRIBUTIONS TO SCIENCE

Indian Contributions to Science : Volume-II